

$$\textcircled{1} \quad y' = y^2$$

(a) Is this a linear ODE?

No. y^2 is not a term allowed in linear ODE.

(b) Find general solution.

$$y' = y^2 \rightsquigarrow \frac{dy}{dt} = y^2$$

$$\rightsquigarrow \frac{dy}{y^2} = dt$$

Integrate both sides: $\frac{-1}{y} = t + C$

$$\Rightarrow \boxed{y(t) = \frac{-1}{t+C}}$$

$$\frac{-1}{t - \frac{1}{2}}$$

(c) Solve IVP with $y(0) = 2$.

$$2 = y(0) = \frac{-1}{0+C} = \frac{-1}{C}$$

$$\Rightarrow C = -\frac{1}{2}$$

$$\text{Solution is } y(t) = \frac{-1}{t - \frac{1}{2}} = \frac{2}{1 - 2t}$$

$$(2) \quad y' - 2y - t = 0 \quad (*)$$

(a) Write down homogeneous version & solve.

$$y' - 2y = 0$$

$$\Rightarrow y' = 2y \Rightarrow \boxed{\text{general solution } y(t) = C e^{2t}}$$

(b) Check $y(t) = -\left(\frac{t}{2} + \frac{1}{4}\right)$ is a solution for (*)

Check: $y' - 2y - t = 0$

$$\left(-\left(\frac{t}{2} + \frac{1}{4}\right)\right)' - 2\left(-\left(\frac{t}{2} + \frac{1}{4}\right)\right) - t$$

$$= -\frac{1}{2} + t + \frac{1}{2} - t = 0 \quad \checkmark$$

(c) Write down general solution to (*).

This ODE is linear.

Just need (general solution to homog. version) + (particular solution)

$$\Rightarrow \boxed{y(t) = C e^{2t} - \left(\frac{t}{2} + \frac{1}{4}\right)}$$

③ Find general solutions:

(a) $2y'' - 4y' + 8y = 0$

Char. equation: $2r^2 - 4r + 8 = 0$

$$\begin{aligned} \text{roots: } r_1, r_2 &= \frac{4 \pm \sqrt{16 - 4 \cdot 2 \cdot 8}}{4} \\ &= \frac{4 \pm \sqrt{-3 \cdot 16}}{4} = \frac{4 \pm 4\sqrt{3} \cdot i}{4} \\ &= 1 \pm i\sqrt{3} \quad \text{complex} \end{aligned}$$

$$\Rightarrow \boxed{y(t) = e^t (A \cos(\sqrt{3}t) + B \sin(\sqrt{3}t))}$$

(b) $y'' - 4y' + 4y = 0$

Char. equation: $r^2 - 4r + 4 = 0$
 $(r - 2)^2 = 0$

repeated root $r = 2$

$$\Rightarrow \boxed{y(t) = A e^{2t} + B t e^{2t}}$$

$$(c) \quad y'' - 9y' + 20y = 0$$

Charac. equation: $r^2 - 9r + 20 = 0$

$$(r-5)(r-4) = 0$$

roots: 5, 4

$$\Rightarrow \boxed{y(t) = Ae^{5t} + Be^{4t}}$$

④ Hooke's Law $mx'' = -kx \quad (m, k > 0)$

(a) Solve IVP with $x(0) = 3, \quad x'(0) = 0$

$$mx'' + kx = 0$$

Charac. equation $mr^2 + k = 0$

$$\Rightarrow r^2 = -\frac{k}{m} \Rightarrow r = \pm \sqrt{-\frac{k}{m}} = \pm \sqrt{\frac{k}{m}} \cdot i$$

(Note: $\frac{k}{m} > 0$)

general sol: $x(t) = A \cos\left(\sqrt{\frac{k}{m}} \cdot t\right) + B \sin\left(\sqrt{\frac{k}{m}} \cdot t\right)$

use initial conditions:

$$3 = x(0) = A \cos(0) + B \sin(0) = A \cdot 1 + 0$$

$$\Rightarrow \boxed{3 = A}$$

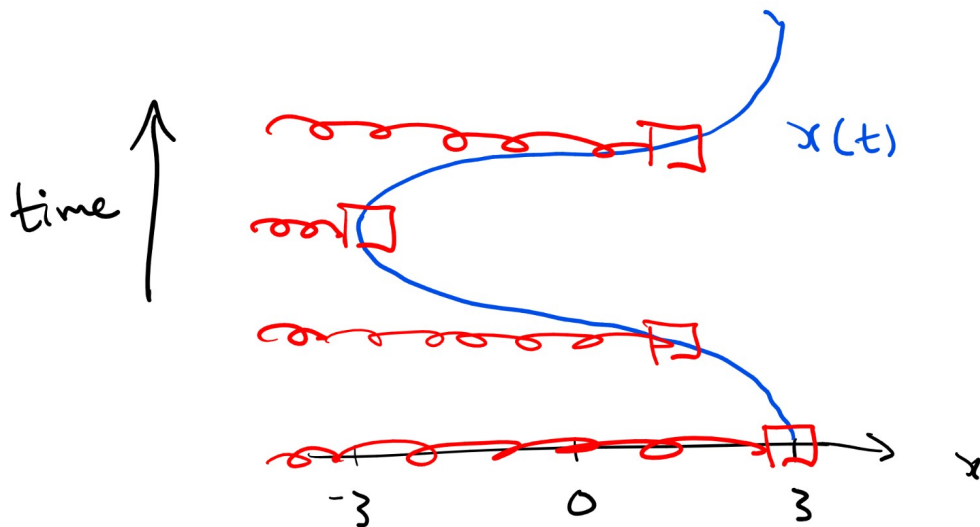
$$x'(t) = -\sqrt{\frac{k}{m}} \sin\left(\sqrt{\frac{k}{m}} \cdot t\right) + \sqrt{\frac{k}{m}} B \cos\left(\sqrt{\frac{k}{m}} \cdot t\right)$$

$$0 = x'(0) = \sqrt{\frac{k}{m}} \cdot B \Rightarrow \boxed{B = 0.}$$

So, solution to IVP is $x(t) = 3 \cos(\sqrt{\frac{k}{m}} \cdot t)$

(b) Distance from equilibrium at time t
 $= |x(t)|$

max $|x(t)|$ is 3 (amplitude of the oscillation)



(c) What is the period of oscillation?
What is the frequency.

In general, $A \cos(\omega t)$

$\Rightarrow$ period is $\frac{2\pi}{\omega}$ since

$$A \cos(\omega(t + \frac{2\pi}{\omega})) = A \cos(\omega t + 2\pi) \\ = A \cos(\omega t)$$

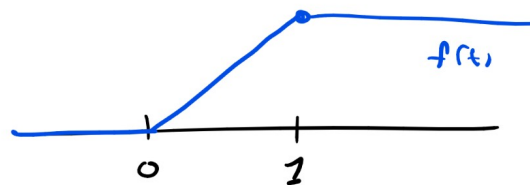
So, period for trajectory is $\frac{2\pi}{\sqrt{\frac{k}{m}}} = 2\pi \sqrt{\frac{m}{k}}$

In general, period = time / oscillation

So, frequency = oscillation / time = $\frac{1}{\text{period}}$

$\Rightarrow$ frequency for traj. is $\frac{1}{2\pi} \sqrt{\frac{k}{m}}$

$$\textcircled{5} \quad f(t) = \begin{cases} 0 & \text{if } t \leq 0 \\ t & \text{if } 0 \leq t \leq 1 \\ 1 & \text{if } 1 \leq t \end{cases}$$



Compute $L[f]$.

$$L[f](p) = \int_0^{\infty} e^{-pt} f(t) dt$$

$$= \int_0^1 t e^{-pt} dt + \int_1^{\infty} e^{-pt} dt$$

$$= \left[-\frac{t e^{-pt}}{p} - \frac{e^{-pt}}{p^2} \right]_0^1 + \left[-\frac{e^{-pt}}{p} \right]_1^{\infty}$$

$$= \left(\cancel{-\frac{e^{-p}}{p}} - \frac{e^{-p}}{p^2} + \frac{1}{p^2} \right) + \underbrace{\left(\lim_{R \rightarrow \infty} -\frac{e^{-pR}}{p} \right)}_{=0 \text{ when } p > 0} + \cancel{\frac{e^{-p}}{p}}$$

$$= \frac{1}{p^2} - \frac{e^{-p}}{p^2}$$

$$= \frac{1}{p^2} (1 - e^{-p}) \quad \text{when } p > 0$$

Scratch by parts

$$\int t e^{-pt} dt$$

$$= -\frac{t e^{-pt}}{p} + \int \frac{e^{-pt}}{p} dt$$

$$= -\frac{t e^{-pt}}{p} - \frac{e^{-pt}}{p^2}$$

⑥ Find $L[\cosh(at)]$, $L[\sinh(at)]$

$$\begin{aligned}L[\cosh(at)] &= L\left[\frac{e^{at} + e^{-at}}{2}\right] \\&= \frac{1}{2} \left(L[e^{at}] + L[e^{-at}] \right) \\&= \frac{1}{2} \left(\frac{1}{p-a} + \frac{1}{p+a} \right) \quad (\text{assuming } p > 0) \\&= \frac{1}{2} \left(\frac{(p+a) + (p-a)}{(p-a)(p+a)} \right) = \frac{1}{2} \left(\frac{2p}{p^2 - a^2} \right) \\&= \frac{p}{p^2 - a^2}\end{aligned}$$

$$\begin{aligned}L[\sinh(at)] &= L\left[\frac{e^{at} - e^{-at}}{2}\right] \\&= \frac{1}{2} \left(L[e^{at}] - L[e^{-at}] \right) \\&= \frac{1}{2} \left(\frac{1}{p-a} - \frac{1}{p+a} \right) \\&= \frac{1}{2} \left(\frac{(p+a) - (p-a)}{(p-a)(p+a)} \right) = \frac{1}{2} \left(\frac{2a}{p^2 - a^2} \right) \\&= \frac{a}{p^2 - a^2}\end{aligned}$$

⑦ Derive an expression for $L[y''']$ in terms of $L[y]$

We know $L[f'] = p L[f] - f(0)$

$$L[f''] = p^2 L[f] - p f(0) - f'(0)$$

$$L[y'''] = L[(y')''] = p^2 L[y'] - p y'(0) - y''(0)$$

$$= p^2 (p L[y] - y(0)) - p y'(0) - y''(0)$$

$$= p^3 L[y] - p^2 y(0) - p y'(0) - y''(0)$$

Note:  degree of p = # of derivatives on y .

⑧ Find $f(t)$ with $L[f] = \frac{4-p}{(p-1)(p+2)}$

partial frac:

$$\frac{4-p}{(p-1)(p+2)} = \frac{A}{p-1} + \frac{B}{p+2}$$

$$\Rightarrow 4-p = A(p+2) + B(p-1)$$

eval. at $p=1 \rightsquigarrow 4-1 = A(1+2) \rightsquigarrow A=1$

eval at $p=-2 \quad 4+2 = B(-2-1) \rightsquigarrow B=-2$

$$\begin{aligned}\frac{4-p}{(p-1)(p+2)} &= \frac{1}{p-1} - \frac{2}{p+2} \\ &= \mathcal{L}[e^t] - 2\mathcal{L}[e^{-2t}] \\ &= \mathcal{L}[e^t - 2e^{-2t}]\end{aligned}$$

$$\Rightarrow \boxed{f(t) = e^t - 2e^{-2t}}$$

⑨ Solve: $y' + 2y + 2 = 0, \quad y(0) = 0$

$$\mathcal{L}[y'] + 2\mathcal{L}[y] + 2\mathcal{L}[1] = 0$$

$$p\mathcal{L}[y] - \cancel{y(0)} + 2\mathcal{L}[y] + \frac{2}{p} = 0$$

$$(p+2)\mathcal{L}[y] = -\frac{2}{p}$$

$$\mathcal{L}[y] = \frac{-2}{p(p+2)}$$

Now, have to "take $\mathcal{L}^{-1}$ "

$$\frac{-2}{p(p+2)} = \frac{A}{p} + \frac{B}{p+2}$$

$$-2 = A(p+2) + Bp$$

$$p=0 \rightsquigarrow -2 = 2A \rightsquigarrow A = -1$$

$$p=-2 \rightsquigarrow -2 = -2B \rightsquigarrow B = 1$$

Clever way:

$$\begin{aligned}\frac{-2}{p(p+2)} &= \frac{p-p-2}{p(p+2)} \\ &= \frac{p}{p(p+2)} - \frac{(p+2)}{p(p+2)} \\ &= \frac{1}{p+2} - \frac{1}{p}\end{aligned}$$

$$\mathcal{L}[y] = \frac{-2}{p(p+2)} = \frac{1}{p+2} - \frac{1}{p} = \mathcal{L}[e^{-2t}] - \mathcal{L}[1]$$

$$= \mathcal{L}[e^{-2t} - 1]$$

$$\Rightarrow \boxed{y(t) = e^{-2t} - 1}$$

Check: $y(0) = e^0 - 1 = 0 \quad \checkmark$

$$y' + 2y + 2 \stackrel{?}{=} 0$$

$$(e^{-2t} - 1)' + 2(e^{-2t} - 1) + 2$$

$$= -2e^{-2t} + 2e^{-2t} - 2 + 2 = 0 \quad \checkmark$$